

EcoLoop

A Circular Economy Marketplace for Materials and Sustainable Products

Corrected Project Document — SE3090 Software Engineering Frameworks

This version reflects the team's clarified client structure: **Flutter mobile application for all platform users** and a **React web application for Super Admins**. Business verification through eROC/ROC is an **admin-side process**; users submit their business details and verification request, while Super Admins perform the actual review and verification.

1. Project Vision

EcoLoop is a circular-economy marketplace that connects people and businesses that have recyclable/reusable materials with people and businesses that need them, while also providing a marketplace for sustainable finished products. The platform is designed around reuse, recycling, responsible buying and selling, and trusted business participation.

- **Mobile app:** used by all normal users and business users for marketplace activities.
- **Web application:** used by Super Admins for administration, moderation, verification and management.
- **Shared backend:** both clients communicate with the same ASP.NET Core Web API.
- **Shared database:** PostgreSQL stores the same users, businesses, listings, orders, transactions and statuses.
- **Business verification:** users request verification; Super Admins review the submitted information and perform the eROC/ROC check.

2. Client Applications

Client	Main Users	Main Responsibilities
Flutter Mobile App	All platform users	Registration/login, business profile, material marketplace, product marketplace, buying/selling, m
React Web App	Super Admins	User/business management, verification review, eROC/ROC checking, category management, li

Important Architecture Rule

The assignment requires React and Flutter to use the same ASP.NET Core Web API, PostgreSQL database, identity, permissions and business rules. Therefore, the fact that the web client is admin-only does **not** mean it is a separate system. It is another client of the same backend.

3. User Roles

Normal User: Can buy and sell materials/products. Can create I HAVE listings, create I NEED requests, browse products, place orders, manage transactions and optionally create a business profile.

Business User: A registered user who has created a business profile. Can operate as a seller/buyer according to the business type. Can request verification and, once approved, receive a verified badge.

Recycling Company: A type of business user, typically interested in buying recyclable materials. Can create I NEED material requests and participate in material transactions.

Sustainable Product Business: A type of business user that can list finished eco-friendly, recycled or upcycled products for sale.

Super Admin: Uses the React web application. Manages users, businesses, verification requests, categories, moderation, platform records and administrative workflows.

4. Four Primary Business Components

1. **Business Hub & Verification** — Who is this business, and has the platform verified it?
2. **Materials Marketplace** — What materials do users/businesses have, and what materials do they need?
3. **Sustainable Products Marketplace** — What finished sustainable products are available to buy?
4. **Transactions & Delivery Management** — What happens after a buyer and seller decide to trade or purchase?

5. Component 1 — Business Hub & Verification

This component manages business identity and trust. Any registered user can create a business profile. However, **the user does not verify the business themselves**. The user submits the required information and a verification request. Super Admin reviews the request through the React web application and performs the eROC/ROC verification process.

User-side features — Flutter

- Create business profile.
- Edit/update business information.
- Upload/replace business documents where required.
- Submit a verification request.
- View verification status and history.
- Resubmit after rejection if permitted.
- View verified badge when approved.

Admin-side features — React Web

- View and search business profiles.
- View pending verification requests.
- Inspect submitted registration/business information and documents.
- Perform admin-side eROC/ROC checking.
- Approve or reject verification.
- Record verification decision/reason.
- Manage verification history and business status.
- Apply/remove the verified badge according to the verification status.

Core CRUD

- **Business Profile:** Create, Read, Update, Delete/deactivate.
- **Business Documents:** Create/upload, Read, Update/replace, Delete.
- **Verification Request:** Create, Read, Update through review/decision, Withdraw where applicable.
- **Verification History:** Create and Read for auditability; historical records should normally not be physically deleted.

Key Business Workflow

Registered User → Create Business Profile → Submit Verification Request → Super Admin Reviews → Super Admin Checks eROC/ROC Information → Approve/Reject → Business Status Updated → Verified Badge Shown if Approved.

Scope note: unless an authorized eROC API is established and approved for the project, describe this as **admin-assisted eROC/ROC verification**, not automatic system-to-eROC verification.

6. Component 2 — Materials Marketplace

This is the core two-sided materials exchange. Users are not restricted to being sellers or buyers. The same user can have both **I HAVE** and **I NEED** listings.

- **I HAVE:** a user/business lists recyclable or reusable material they currently have available.
- **I NEED:** a user/business posts a material requirement they want to obtain.

- Examples: PET bottles, plastic, paper, cardboard, glass, metal, textiles, wood and e-waste.
- Recycling companies can primarily use I NEED to announce materials they are looking to purchase/obtain.
- Users can search, filter, sort and paginate material listings/requests.
- A buyer and seller can proceed from a listing/request into the Transactions & Delivery component.

Core CRUD

- **Material Listings (I HAVE):** Create, Read, Update, Delete/close.
- **Material Requests (I NEED):** Create, Read, Update, Delete/close.
- **Material Categories:** Create, Read, Update, Delete (primarily administered by Super Admin).
- **Material Images:** Create/upload, Read, Update/replace, Delete.

Business-specific operations

- Search/filter/sort/pagination
- Respond to or make an offer on a listing/request
- Match I HAVE listings with I NEED requests
- Mark a listing as sold/fulfilled/closed

7. Component 3 — Sustainable Products Marketplace

This is a conventional marketplace for finished eco-friendly, recycled, reused or upcycled products. Verified businesses can sell products, while users can browse and purchase them.

- Examples: recycled bags, recycled paper products, upcycled furniture, reusable products and other sustainable goods.
- Product pages show product information, seller/business information, price, images, stock and availability.
- Users can search, filter, sort and paginate products.
- Users can add products to a cart and proceed to an order.

Core CRUD

- **Products:** Create, Read, Update, Delete/disable.
- **Product Categories:** Create, Read, Update, Delete.
- **Product Images:** Create/upload, Read, Update/replace, Delete.
- **Inventory/Stock:** Create, Read, Update, Delete/disable.

Business-specific operations

- Product search/filter/sort/pagination
- Cart management
- Wishlist/favorites
- Stock management
- Product reviews/ratings
- Show verified seller/business information

8. Component 4 — Transactions & Delivery Management

This component manages the operational process after a user decides to buy or exchange something. It connects marketplace activities with orders, transactions, delivery choices and status tracking.

Material transaction flow

Buyer interest/request → Seller accepts/rejects → Transaction created/confirmed → Ready for pickup or delivery → Completed / Cancelled.

Product order flow

Cart → Order placed → Confirmed → Processing → Ready → Delivered/Completed or Cancelled.

Delivery model

- There is **no EcoLoop-owned delivery rider/fleet**.
- When creating a material/product listing, the seller indicates whether seller delivery is available.
- If seller delivery is available, the buyer chooses **Self Pickup** or **Seller Delivery**.
- If seller delivery is not available, the buyer gets **Self Pickup only**.
- Google Maps services can be used for location, distance and route-related functionality.

Core CRUD

- **Material Transactions:** Create, Read, Update status/details, Cancel.
- **Product Orders:** Create, Read, Update status, Cancel.
- **Deliveries:** Create, Read, Update status/location, Cancel where appropriate.
- **Delivery Locations:** Create, Read, Update, Delete.

Business-specific operations

- Accept/reject transaction
- Choose delivery method
- Calculate route/distance using maps services
- Track transaction/order status
- Maintain transaction/order history
- Complete or cancel a transaction/order

9. High-Level Database Entities

Identity & business: Users, Roles, BusinessProfiles, BusinessDocuments, BusinessVerification, VerificationHistory

Materials: MaterialCategories, MaterialListings, MaterialRequests, MaterialImages

Products: ProductCategories, Products, ProductImages, Inventory

Transactions: Transactions, TransactionItems, Orders, OrderItems

Delivery: Deliveries, DeliveryLocations, StatusHistory

Supporting: Notifications, Reviews/Ratings, Audit-related records where required

10. API & Integration Architecture

- ASP.NET Core Web API is the mandatory public backend.
- Entity Framework Core with the PostgreSQL provider is used for data access.

- PostgreSQL is the shared relational database.
- Flutter communicates with the ASP.NET Core API.
- React communicates with the ASP.NET Core API.
- Neither client should bypass the API to directly access PostgreSQL.
- JWT authentication and role-based authorization should protect user and admin operations.
- Google Maps is the planned third-party integration for location/distance/route functionality.
- If a Python Agentic AI service is used, it operates as an internal service called by ASP.NET Core rather than directly by Flutter or React.

11. Web vs Mobile Feature Boundary

Feature	Flutter Mobile — All Users	React Web — Super Admin
Register/Login	✓	Admin login
Business profile	✓ Create/manage	✓ View/manage
Verification request	✓ Submit/view status	✓ Review/approve/reject
eROC/ROC checking	—	✓ Admin process
Material I HAVE	✓	✓ Admin moderation/management
Material I NEED	✓	✓ Admin moderation/management
Products	✓ Browse/buy; eligible business users manage own products	✓ Admin management/moderation
Orders/transactions	✓ User-facing	✓ Oversight/management
Delivery	✓ User-facing choices/status	✓ Oversight
Categories	View/use	✓ Manage
Reports/administration	—	✓

12. Assignment Alignment

The assignment specifies a four-student group with four primary business components, with each student owning one component and providing evidence across backend, database, React, Flutter, tests, Git, documentation and a distinct Agentic AI contribution. ■filecite■turn9file0■L37-L58■

The assignment also requires the React and Flutter applications to use the same ASP.NET Core Web API, PostgreSQL database, user identity, permissions and business rules. ■filecite■turn9file1■L76-L92■

The required stack includes ASP.NET Core Web API, PostgreSQL/EF Core, React and Flutter, with GitHub and GitHub Actions CI and suitable testing. ■filecite■turn9file0■L17-L35■

The specification also requires an end-to-end workflow that can cross clients and pass through ASP.NET Core, PostgreSQL and Agentic AI, with review/approval in the other client before the updated status is returned. ■filecite■turn9file3■L190-L195■

13. Four-Person Ownership Structure

Student	Primary Component	Main Domain Ownership
Student 1	Business Hub & Verification	Business profiles, documents, verification workflow, admin review and verified status.
Student 2	Materials Marketplace	I HAVE, I NEED, material categories/listings/requests and material matching.
Student 3	Sustainable Products Marketplace	Products, categories, images, inventory and product shopping features.
Student 4	Transactions & Delivery Management	Orders, material transactions, delivery choices, status workflows and maps integration.

Important: primary ownership does not mean working only on one layer. The assignment explicitly expects each student to demonstrate work across backend, database, React, Flutter, testing, Git, documentation and a distinct Agentic AI contribution. ■filecite■turn9file0■L43-L58■

14. Scope Boundaries

- EcoLoop is not a garbage collection service.
- EcoLoop does not need its own delivery rider network.
- Business verification is performed by Super Admins; users only submit information and request verification.
- The React web application is not a second marketplace client for general users; it is the administration client.
- The Flutter application is the main client for all platform users.
- The four business components should remain distinct but integrated.
- Agentic AI responsibilities are intentionally not finalized in this document; they can be designed after the core business workflows are stable.

15. Overall EcoLoop Flow

User registers → uses Flutter mobile app → creates/uses personal or business profile → optionally requests business verification → Super Admin reviews through React web → verified business badge appears → users/businesses post I HAVE or I NEED materials → users browse sustainable products → buyers initiate transactions/orders → seller accepts/processes → buyer chooses Self Pickup or Seller Delivery where available → Google Maps supports location/route functions → transaction/order is completed.

16. Assignment Checklist to Keep in Mind

- Four primary business components for four students.
- ASP.NET Core Web API + PostgreSQL + EF Core.

- React admin web application.
- Flutter mobile application for all users.
- JWT authentication and role-based authorization.
- Meaningful CRUD and specialized business operations for each component.
- At least four specialized agents with controlled tools and structured state.
- Validation, observability and human approval where required.
- Meaningful third-party integration.
- Traditional testing, Agentic AI evaluation and performance testing.
- GitHub Actions CI.
- ADR and documentation.
- Deployment of React, ASP.NET Core and PostgreSQL, plus Flutter APK generation.

Reference: SE3090 Assignment 1 — Integrated Full-Stack and Agentic AI Application Development, 2026. The project-specific EcoLoop design above is the team's proposed interpretation and organization of the assignment requirements; it is not itself part of the official assignment specification.